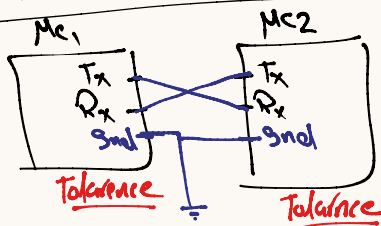


UART

- * Universal Asynchronous Receiver Transmitter
- * Communication Protocol $\rightarrow$ Not standard.
- * need know $\rightarrow$ ① H.W Interface
② Spec.
③ Frame

① H.W Interface

* external H.W Interface



2 wire data Connection

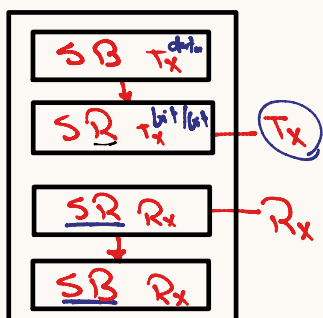
* Need Band Rate

* same Gnd

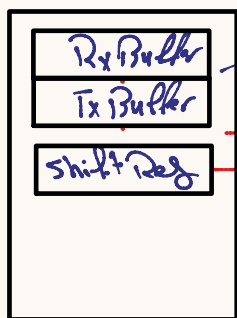
* Internal H.W Interface

* SB Serial Buffer

* SR Shift Register



In Atmega32



$UDR = 8 \text{ bit}$
 $\rightarrow UDR = \text{data}$
 $\rightarrow Tx \text{ Buffer}$
 $R = UDR;$
 $\rightarrow Rx \text{ Buffer}$

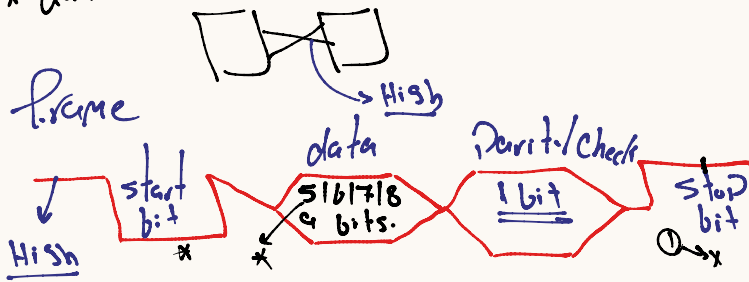
② Specs

Uart

- ① Medium $\Rightarrow$ wire
- ② Transmission technique $\Rightarrow$ serial
- ③ Rx/Tx Relation $\Rightarrow$ Peer to Peer
- ④ stnch / As-nch $\Rightarrow$ As-nch (Baud Rate)
- ⑤ Data Direction $\Rightarrow$ full duplex
- ⑥ throughput $\Rightarrow$
 $\rightarrow$ Frame

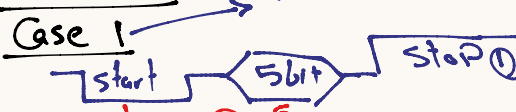
③ Frame Format

* data wire $\rightarrow$ Automatic ideal state is High



- ① start bit $\rightarrow$ 1 bit low
- ② stop bit $\rightarrow$ 1 bit High
- ③ data bits $\rightarrow$ 5:9 bits
- ④ Parity bits $\rightarrow$ 1 bit

throughput $\rightarrow$ min data | min frame



No of Frame $\rightarrow$ 7 bit
No of data $\rightarrow$ 5 bit

$$\text{throughput} = \frac{5}{7} \times 100 \Rightarrow 71\%$$

Case 2 $\rightarrow$ min data $\rightarrow$ Max frame



No of Frame $\rightarrow$ 1 bit
No of data $\rightarrow$ 5 bit

$$\text{throughput} = \frac{5}{9} \times 100 \Rightarrow 55\%$$

Case 3 $\rightarrow$ Max data | Min frame



No of Frame $\rightarrow$ 11 bit
No of data $\rightarrow$ 9 bit

$$\text{throughput} = \frac{9}{11} \times 100 \Rightarrow 81\%$$

Case 4 $\rightarrow$ Max data | Full frame



No of Frame $\rightarrow$ 13 bit
No of data $\rightarrow$ 9 bit

$$\text{throughput} = \frac{9}{13} \times 100 \Rightarrow 69\%$$

Min throughput $\rightarrow$ 55%

Max throughput $\rightarrow$ 81%

* Parity Check

↳ kind of check error mechanism

data

↳ Count ones in frame at Rx

Parity Config → if count is even → correct
if count is odd → Not X

	start	data	P	stop		Rx
①	0	11110110	0	11	→	0 1111 0110 011 ✓
②	0	11100110	1	11	→	0 1110 0110 111 ✓
③	0	11110010	1	11	→	0 1111 0011 111 ✗
④	0	11110010	1	11	→	0 1111 0111 111 ✓

if Parity check Config as even

↳ checking correct when the noise effect on the odd bits

↳ Can't checking correct when the noise effect on the even bits

if Parity check Config as odd

↳ checking correct when the noise effect on the even bits

↳ Can't checking correct when the noise effect on the odd bits

* Speed i. Baud rate

↳ s - 1 Mbit/sec

bit byte frame word

→ Baud rate = bit rate = No. of bits in sec

